ARAVALLI BUILD CHAMPIONSHIP — ROAD TO DEVFEST UDAIPUR • HERITAGE TRACK

"Virasat Edge" Master Plan & Architecture

Integrating Spatial Hardware Sensing, Static BLE Beacons, and Flutter App Wireflows

1. Spatial Edge Sensing & Proximity Workflow

HARDWARE ARCHITECTURE

Virasat Edge deploys privacy-first ESP32-CAM nodes inside historic heritage lanes. Operating at a 4–6 ft field of view (optimized for 180–190 cm height angles), the on-device Edge Impulse model calculates pedestrian crowd count integers without saving video frames, while static BLE beacons broadcast unique identifiers.

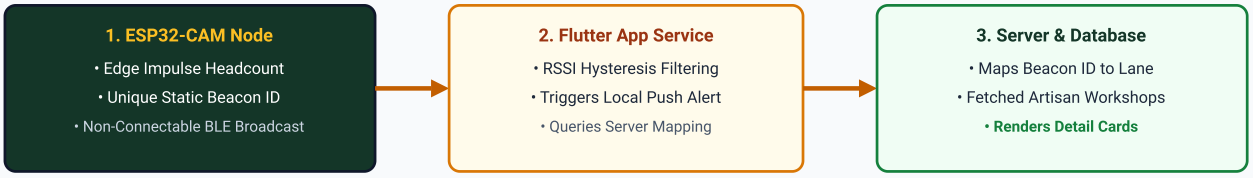


Figure 1: Hardware Edge Sensor to Mobile Proximity Notification Workflow.

2. Mobile App Workflow & Proximity Zones

FLUTTER APP UX

- **Overlapping Beacon Zones:** As tourists move through overlapping BLE coverage areas (Beacon A and Beacon B), the Flutter background service calculates real-time RSSI signal strength.
- **Proximity Notification:** Once the signal passes the entry threshold, the phone fires a clean local notification: *"You are near [Lane Name] — Tap to explore verified local artisans!"*
- **Listing & Distance UI:** Tapping the alert opens bottom-sheet listings displaying artisan titles, craft categories (*Food, Handloom, Potters*), and exact distance markers.

3. Server Mapping Schema & Review System

BACKEND ARCHITECTURE

Module / Component	Key Entities & Attributes	Functionality & Purpose
Static Beacons	`beacon_id`, `lane_id`	Maps physical ESP32 hardware identifiers directly to specific heritage alleyways on the server.
Artisan Details	`name`, `Est. since`, `bio`, `images`	Renders detailed artisan profiles, craft history, and specifications when a user clicks a listing.
Directions & Navigation	`Google Maps Intent`	Provides direct action buttons that hand off coordinates to Google Maps when tourists request walking directions.
Review & Anti-Bot Filter	`Verified Presence`	Validates tourist reviews against physical presence to eliminate fake click-farm ratings.
Firebase Prototype	`Realtime Sync`	Powers live crowd density counters and artisan catalog synchronization.

4. Core Features & Hackathon Execution Focus

FEATURE CHECKLIST

- **Inter-communication:** Seamless synchronization between ESP32 hardware sensors, mobile app background service, and cloud database.
- **Fake Review Filter:** Protecting authentic third-generation artisans from manipulated public map scores.
- **Saved Favorites:** Allowing tourists to bookmark artisan workshops for later visits.

> **Hackathon Pitch Summary**
>
> *"Virasat Edge combines privacy-first ESP32-CAM crowd counting with static BLE proximity beacons and server-mapped artisan directories—giving tourists a seamless way to discover hidden heritage craftsmen while completely eliminating fake review bots."*